What Prediction Leaves Out: Relational Objecthood and the Grounds of Purpose

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Abstract

When does a successful representation of a system justify claims about what persists and what its activity is for? This paper reconstructs Universal Evolutionary Object Theory around that question. It defends a relational realism about organized processes and a conditional account of purposive explanation, while withdrawing the original programme's universal loop requirement and unrestricted optimization claims. Its central argument concerns preservation: transferring evidence from prediction to persistence or purpose requires retaining the distinctions needed for the intended attribution, at its stated level of precision. A finite regenerative model makes this requirement concrete. The same four initial configurations admit two, three, or four predictive classes as the recorded task changes; states indistinguishable by their complete birth records can differ in repair expenditure. In the same source, population growth can favour an action that reduces the original individual's survival, and an optimal policy need not be an internally executed programme. These results neither establish an ontology nor naturalize purpose by themselves. They identify where causal implementation, historical continuity, feasible control, and independently supported evaluation must enter. Comparison with computational mechanics, process philosophy, and organizational teleology locates the proposed contribution in their constrained integration, rather than in priority for their constituent ideas. The resulting unity is a defeasible discipline of explanation, with explicit conditions under which apparently compatible accounts fail to explain the same organized system.

Keywords: objecthood; predictive equivalence; persistence; teleology; organizational realism; UEOT

1 Introduction: two questions about one system

Consider a device that loses a working instruction and restores it from surviving internal copies. Compare it with a device whose working instruction was never damaged. Suppose the two subsequently produce exactly the same distribution of observable outputs. For predicting those outputs, remembering the damage may be unnecessary. For reconstructing this device's actual repair expenditure, the distinction is indispensable. Now suppose an operating mode increases the expected number of descendants but slightly reduces the original device's chance of remaining intact. Describing that mode as promoting the system's persistence is ambiguous. It matters whether the system means this individual, the organization reproduced in its successors, or a population counted at a later date.

These are questions about the entitlement to transfer an explanation. An adequate account of outputs can leave out a real difference in the conditions of their production. An adequate account of reproduction can leave out the death of the reproducing individual. Neither account becomes false because it is selective. Trouble begins when its success is taken to establish a stronger conclusion about the object or its purpose. The central claim of this paper is that object identification and purposive attribution are distinct but mutually constraining explanatory tasks. A representation adequate for one supports the other only if, with any justified supplementary evidence, it preserves what the intended attribution

requires. Exact individual histories, probability laws, and aggregate expectations can require different information.

Universal Evolutionary Object Theory, or UEOT, originally approached these questions through a much stronger programme. It proposed that every persistent object depends on an internal self-maintaining loop, that change reflects physical and informational drives, and that a globally optimal direction and attractor could provide a naturalistic basis for teleology. The original paper also described itself as a research programme and contained qualifications about innovation and changing environments. Those qualifications did not remove its explicit universal assertions (Qian, n.d.). The present article retains the question of how objects and purposes belong in one naturalistic account while revising the proposed answer.

The revision is substantive. Persistence does not generally entail a topological cycle. Predictive equivalence does not establish numerical identity. An optimal-control calculation does not show that an organism executes the calculated policy. Growth of a population is not survival of an individual. Nor does successful maintenance, however accurately described, establish an obligation that the maintained organization ought to continue. The current UEOT materials acknowledge these separations and provide conditional positive results, explicit counterexamples, and a finite regenerative source in which several explanatory relations can be checked together (Qian, 2026b,a). This article develops a philosophical argument from those distinctions. It is not a summary of the books or a claim that a proof assistant has established a metaphysics.

The two positions defended here require different arguments. The first is a selective relational realism: some organizations are real because their relations support stable causal explanations across appropriately varied encounters, even though no single predictive partition captures everything about them. The realist commitment is defeasible and extends beyond the mathematics of prediction. The second is a conditional account of purposive explanation: a claim that activity is for an outcome requires evidence connecting a relevant evaluative distinction with the system's implementation or history. Such evidence can concern regulatory control, a selected effect, or a designed function. These grounds are not interchangeable, and they do not require a unique numerical utility.

Their connection is the paper's principal contribution. A purpose belongs to a specified bearer or organized activity. Changing the bearer from an individual to a lineage may change the apparent purpose without changing the underlying events. Conversely, attributing a function can reveal distinctions that an otherwise adequate predictive representation must retain. The question is therefore not simply which ontology is predictive or which objective is optimal. It is whether the representation of the bearer and the explanation of its activity can be satisfied by the same process under the same restrictions. Separate examples of a repairing system and an optimizing system do not establish that one admissible system does both; a common realization is an additional achievement.

The ingredients have substantial precedents. Computational mechanics develops predictive equivalence and minimal representations. Process philosophy understands persistence through organized change. Organizational accounts of function and agency already connect individuality, regulation, and normativity (Shalizi and Crutchfield, 2001; Dupré and Nicholson, 2018; Mossio et al., 2009; Barandiaran et al., 2009). Recent work explicitly discusses regulatory teleology and the persistence of the organized whole (Bich, 2024; Cuciniello, 2026). The claim here is consequently narrower than a new foundation for all these positions. It is an argument about their compatibility, sharpened by paired failures and a restricted positive construction in one mathematical source. Its value depends on whether that construction exposes explanatory errors that would otherwise remain easy to overlook.

The argument proceeds from identification to persistence and then to purpose. Section 2 explains which parts of the original programme require revision. Section 3 distinguishes a predictive type from an enduring bearer and develops the preservation condition. Section 4 examines organization, repair, and historical continuity. Section 5 shows why purposive explanation needs more than optimization, using exact growth results from the same source. Section 6 explains the limited unity achieved and its relation to the closest alternatives. Section 7 addresses objections to the philosophical position and its evidential reach.

2 Reconstructing the original programme

2.1 What survives revision

The enduring motivation of UEOT is that objects should not be treated as inert bearers whose changes are philosophically secondary. In many explanations, what persists is an organized capacity sustained through material or informational replacement. The original programme asked whether this perspective could also explain why some activities appear directed toward maintaining or transforming their conditions of existence. Those questions survive the failure of a universal answer. A research programme can retain a problem while abandoning premises that previously made its solution look immediate.

Three commitments are worth separating. First, relational organization can be explanatorily important: what a component does depends on its connections, available resources, and the larger process in which it operates. Second, temporal maintenance is a phenomenon requiring explanation, not a primitive guarantee supplied by naming an object. Third, purposive language should be accountable to natural processes. None entails that every object regenerates itself, that every persistent system is an agent, or that nature possesses one privileged objective. A stone, a repaired register, a reproducing lineage, and an autonomous organism can share some explanatory features while differing in others.

The original distinction between Type I and Type II objects needs a fair reading. It was presented as a contrast between state-dependent organizational tendencies, not simply an immutable division of the world into two kinds (Qian, n.d., 4-5). Even so, a binary vocabulary is too coarse to settle whether a particular process merely endures, repairs damage, regulates a variable, reproduces its organization, or acts with autonomous agency. The present account replaces reliance on that contrast with explicit claims about these capacities. It does not infer the strongest capacity from the weakest.

2.2 What must be withdrawn or narrowed

The original loop postulate included nontrivial first homology of a state space or interaction network. This is stronger than saying that a persisting system has causal dependencies. A contracting flow, such as $\dot{x} = -x$ on a contractible state space, can remain in a bounded region and approach a stable fixed point without nontrivial first homology of that space. Stable state dynamics therefore do not entail this state-space topology (Qian, 2026a, §§8-9). The example does not exclude a cyclic interaction network in some realization of that flow; the original paper explicitly allowed that alternative (Qian, n.d., 4). A universal claim about an object's network or regenerative organization needs a separate realization argument. It cannot be inferred from state stability, and this flow alone does not refute every such alternative. The reconstruction withdraws the universal requirement because it lacks that general justification, while retaining specific organizational hypotheses where they are supported.

The physical and informational quantities denoted by Π and Φ also require separate measurement and interpretation. If only a combined expression $J = \Pi - \lambda\Phi$ is observed, the transformation

$$\Pi' = \Pi + \lambda\chi, \qquad \Phi' = \Phi + \chi$$

leaves J unchanged. The decomposition is not identified by the combined score alone. In the original discussion, informational potential also shifted between uncertainty, adaptive novelty, and evaluative significance. These uses cannot be unified by notation. The current sources further supply a local obstruction to a universal smooth two-dimensional representation. For the healthy care programme introduced below, the public probabilities of retention, birth, and the joint event of birth with damaged output are (s, b, bq). Holding δ, γ fixed, their Jacobian with respect to (s, b, q) has determinant $b > 0$ on the legal open domain (Qian, 2026a, §23.4). These responses cannot all factor there through one common smooth two-dimensional intermediate representation. This is model-internal parameter variation, not a demonstrated physical experiment, and it does not exclude every discrete or set-theoretic encoding.

The language of a Global Optimal Direction and a Global Optimal Attractor must likewise become conditional. An optimum is relative to an objective, feasible actions, information, resources, and a time criterion. Optimal actions can tie. An invariant distribution, a conditional distribution among survivors, a deterministic limit set, and a family of optimal-policy limits are different mathematical objects. The current theory has results about particular members of these categories; it does not establish a unique attractor toward which every natural system progresses (Qian, 2026a). The original paper's discussion of innovation already made an unchanging universal optimum problematic, but its explicit convergence claims still require withdrawal rather than reinterpretation as harmless metaphor.

Finally, the original extension from alignment with a global direction to moral obligation does not survive the reconstruction. A mechanism's conditions of operation can support judgments of functional success or failure. They do not by themselves establish who ought to preserve it, whether its continuation is desirable, or whose costs count. The new article leaves moral justification to further argument. This is a change in the scope of the theory, not merely a change of terminology (Qian, n.d., 26-28; Qian, 2026b).

These revisions leave a positive task. If universal loops and objectives cannot supply identity and purpose in advance, one must investigate the relations that make a particular attribution defensible. The rest of the paper argues that these relations constrain each other. A formal distinction is useful here precisely because it can prevent an attractive philosophical interpretation from acquiring more support than its premises provide.

3 Predictive identity and the reality of organization

3.1 A representation answers a question

Let a history h determine response laws $K_i(h, \cdot)$ for declared protocols i. Its canonical predictive representation is the family

$$C(h) = \big(K_i(h, \cdot)\big)_i.$$

Histories agree predictively when they yield the same relevant response laws. Under the measurable and countability assumptions of the current UEOT result, any statistic through which all these laws factor

also determines C almost everywhere with respect to the specified reference measure. The conclusion is a precise minimality statement. It is not the assertion that every conceivable conditional probability has a pointwise canonical value, including on unobserved null histories (Qian, 2026a, §§4.1-4.3; UEOT Project, 2026c). The supplementary formal matrix supplies the exact source route as F01; the common-source construction discussed next is F02.

This construction belongs to an established mathematical tradition. Causal states identify histories by their distributions of futures; transducer constructions extend predictive equivalence to input-output processes (Shalizi and Crutchfield, 2001, causal-state definition and theorems 1-3; Barnett and Crutchfield, 2015, causal-equivalence and optimality results). Conditioning on supplied inputs does not by itself establish causal intervention semantics. The current UEOT source supplies its own declared protocol and noise restrictions, and connects certain response laws to a common latent process. That connection matters: a conditional future must arise from the process that generated the past, rather than from an independently substituted hidden state that changes the conditional joint law. Correct posterior resampling is not excluded. But neither minimality nor source consistency changes the kind of identity initially established. It is equivalence for a specified response task.

The investigator chooses which question to ask. Once that question and the model are fixed, its answer is constrained. A representation that merges histories with different required future distributions is inadequate, even if it is simpler or philosophically attractive. Conversely, adding a historical distinction without predictive consequences may make a statistic redundant for that task. These are objective relations among specified models and questions. They do not yet determine which questions reveal an enduring object.

Calling predictive classes types helps avoid a premature ontological inference. Two numerically distinct devices can instantiate the same predictive type. One device can move between predictive types after damage. A coarser task can merge states that a richer task separates. None of these facts implies that the device has ceased to exist, that two devices have become one, or that objecthood is unreal. It implies that identifying a response type and tracking a bearer are different operations.

3.2 One source, three legitimate partitions

The finite regenerative source makes the dependence on the task exact. There are two internal programmes, $\theta_0 = (\text{care}, \text{care})$ and $\theta_1 = (\text{bloom}, \text{care})$, and two initial conditions, healthy H and damaged D. The ordered programme entries specify actions after a healthy or damaged diagnosis. Bloom trades some individual retention for increased birth probability; care is the affordable executable action after damage and paid repair. Let B_t count strict births across the whole actual population at census t, excluding every retention event. A newborn first executes at the following census. The four initial configurations are $\theta_0 H$, $\theta_0 D$, $\theta_1 H$, and $\theta_1 D$. Under Section 5's probability conditions together with $\gamma > \delta$ and $s + b > 1$, the birth advantage exceeds the retention loss and care has expected total output above one. In this original positive regime, the partitions are those in Table 1 (Qian, 2026a, opening definitive argument, part III).

Recorded task	Predictive classes of the four initial configurations
First strict-birth count B_1	$\{\theta_0 H, \theta_0 D, \theta_1 D\}$; $\{\theta_1 H\}$
Joint birth records $(B_1, \ldots, B_T)$ for every finite T	$\{\theta_0 H, \theta_0 D\}$; $\{\theta_1 D\}$; $\{\theta_1 H\}$
Those complete finite birth records together with first repair fee R_1	Four singleton classes

Table 1. Task-dependent predictive distinctions within one fixed-programme source. The result is a written proof with independent exact verification in the source record; it is not represented here as a complete dedicated Lean theorem. It concerns these fixed internal programmes, not the entire class of external control policies.

The healthy and damaged starts of θ_0 can be coupled to have the same entire birth record after the damaged body is restored. Their initial repair expenditures differ: zero for H and three units for D. The complete birth process consequently omits a distinction relevant to the explanation of resource use. At the same time, extending the first-step birth question to all finite joint birth records separates $\theta_1 D$ from $\theta_0 D$. Their initially identical care actions lead to descendants that carry different programmes and can act differently when healthy. The distinction appears in the second census. At $(s, \delta, b, \gamma, q) = (3/4, 1/8, 3/8, 1/4, 1/3)$, the expected second birth counts for $\theta_0 D$ and $\theta_1 D$ are $27/64$ and $39/64$, respectively.

The third row requires attention to the whole task. Merely recording (B_1, R_1) produces three classes, since both damaged programmes initially undergo the same payment and care action. Four classes result when the repair fee is added to the complete finite birth-record task. This detail is philosophically instructive. The record that resolves one ambiguity need not resolve another; adequacy belongs to the joint question actually posed.

All three partitions can be correct. The first does not falsify the third, and the third does not reveal that the first was secretly describing four predictive objects. They identify different equivalence relations in the same source. Yet only the richer representation can carry an explanation that explicitly distinguishes a paid recovery from undamaged operation. If one uses the coarser birth representation to argue that repair costs are constitutively irrelevant to the organization, the inference fails. What was irrelevant to a selected output task has been promoted into an unrestricted claim about the process.

3.3 The preservation condition

The logical point is elementary but consequential. Suppose $q : S \to Z$ is a representation of source situations and F is a feature required by an explanation. A well-defined feature $\bar{F}$ of the represented states can satisfy $F = \bar{F} \circ q$ only if F agrees whenever q agrees. Conversely, in the simple set-theoretic case, agreement on every fibre of q defines such a $\bar{F}$ on its image. If the explanation concerns histories, the same question must be asked of the history representation, not just the instantaneous state map. For measurable or probabilistic claims, the relevant regularity and almost-everywhere qualifications must also be retained.

No new mathematical theorem is claimed by this observation. Its immediate scope is an exact attribution of F from the stated representation alone. A probabilistic or aggregate explanation need not preserve each realized individual value. Suppose a quotient merges repair fees zero and three, and independent evidence supplies equal probability for the two cases. The quotient together with that evi-

dence can determine a fee distribution and expected fee $3/2$, although it cannot identify which fee this bearer paid. For a distribution, expectation, or approximation, preservation concerns that stated target; supplementary evidence belongs in the combined representation being assessed.

The philosophical proposal is to use this distinction as a discipline on explanatory transfer. Identify the bearer, explanatory contrast, temporal scope, and basis of evaluation before claiming that a convenient representation answers the question. If an exact persistence claim requires identifying whether a bearer survived, the available evidence must retain that distinction or supply a justified history map. If an explanation identifies a particular paid restoration, subsequent outputs alone can be insufficient. If optimal-action reasoning relies on affordability, an observation class erasing the relevant difference cannot support that same state-specific decision. These restrictions allow useful aggregate explanations while blocking their unsupported promotion to individual histories.

The condition is necessary, not an automatic warrant. Constant features on a quotient can still be interpreted badly, the source model can be false, and an evaluative distinction can be arbitrary. Preservation prevents information already known to matter from being silently lost. It does not manufacture the evidence that the information is true or relevant. In particular, choosing the conclusion as a new feature would make it formally preserved while doing nothing to justify it. The burden is to establish the feature's role independently and then determine whether the representation retains it.

There is a positive counterpart in the same source. A causally silent physical tag can be erased while preserving the declared projections of transactions, admissible observations, payments, and marked histories. Raw bodies carrying different tags are not literally identical; the relevant records and history laws agree under the source's projection. Erasing the diagnosis can instead remove a distinction in affordability, while preserving decoded instruction bits alone can lose the distinction between an open and a closed execution gate (UEOT Project, 2026c). The principle therefore does not demand maximal detail. It distinguishes justified omission from omission that changes the target explanation.

3.4 From representation to selective realism

Prediction alone does not deliver realism. The further argument is that an organization deserves realist interpretation when its relations explain stable differences across independently specified perturbations, its realization can be traced, and competing representations can be corrected by the same worldly process. The grounds are causal and comparative. In the regenerative example, whether the gate is open and whether the backups survive make differences to actual execution and repair. These relations are not made true by preferring one vocabulary. In a natural case, analogous claims would require observation and intervention rather than inspection of stipulated source code.

This position is close to pattern realism without reducing all reality to compression. Dennett's account already connects predictive patterns and ontological commitment, while allowing that more than one useful pattern can be found (Dennett, 1991). The present argument concerns the extra entitlement needed for a particular persistence or purpose claim. It also respects the difference between structural knowledge and a stronger ontological thesis emphasized in debates about structural realism (Ladyman, 1998). A mathematical quotient cannot settle that debate by changing its name to an object.

Protocol relativity thus need not imply that organization is an invention. Selecting the temperature to measure does not select its value; selecting a response task does not select which histories are equivalent. Nevertheless, a modeler can make a preferred classification trivially correct by choosing a weak enough task. The realist inference gains strength when the organization survives relevant changes in measurement, when failed interventions are recorded, and when the model risks being wrong. The result

is pluralism under constraint, rather than a universal algorithm for identifying the one true partition of nature.

4 Persistence, repair, and the bearer of a history

4.1 Organization is more specific than recurrence

Rejecting a universal topological loop leaves several distinct explanatory possibilities. A dynamical system can remain within a domain without repairing itself. A controller can return a variable to a range without producing its own components. A network can regenerate components while depending on external energy and trusted constraints. These distinctions concern what actually sustains the capacity, not merely the vocabulary used to describe repeated activity. A trajectory's recurrence, a graph cycle, feedback in a controller, and closure among constraints are therefore not interchangeable evidence.

The organizational approach offers a stronger biological hypothesis than generic recurrence. Autopoiesis concerns a productive organization that generates its components and realizes a unity; reproduction is not simply substituted for that organization (Varela et al., 1974). Closure of constraints concerns interdependencies among relatively stable constraints on material and energetic processes, with attention to their timescales (Montévil and Mossio, 2015, definitions 1-4 and §§5-6). The contracting-flow example in Section 2 neither instantiates nor refutes those hypotheses. It separates stable state dynamics from nontrivial state-space homology. An externally driven cycle can also repeat without regenerating its driver, while a regulator can maintain a capacity without following a closed geometric orbit. Thus trajectory topology, network feedback, and regenerative dependence answer different questions. Which organization a living system realizes needs its own evidence.

The revised use of Ω illustrates both the gain and the limit of a more precise approach. Fix a persistence certificate, a horizon, a family of permitted deletions, and a failure criterion. When failure is monotone under deletion, its finite destructive sets can be reconstructed from their minimal members. These minimal failures can identify dependencies relevant to the certificate (Qian, 2026a, §12.2; UEOT Project, 2026c, supplementary matrix F05-F07). The construction depends on those choices and on monotonicity. Removing an inhibitor may activate a backup, so an arbitrary deletion semantics need not be monotone. A restricted intervention family can also have no destructive member at all.

This causal reading of Ω cannot quietly inherit the original universal claim. If the permitted interventions remove only an irrelevant input, their failure family says little about the robustness of the remaining organization. Conversely, losing access to a readout can prevent certification while leaving the process intact. A useful dependency analysis must distinguish loss of evidence from loss of the mechanism itself. The revised concept is valuable when its intervention and certification conditions are informative; it is empty when those conditions have been chosen merely to secure a desired verdict.

4.2 A repair that is implemented rather than stipulated

The regenerative source supplies a modest positive case. Each body has two instruction rows, with one active bit and two backup bits per row, a main execution gate, and a bootstrap capability. The specified damage flips the active bits and closes the gate while retaining the backups. The repair routine reads the available copies, restores the active bits by majority, and reopens the gate. Subsequent execution reads the restored physical cells in the model. It does not consult an externally supplied label declaring the correct programme (UEOT Project, 2026c).

Resources are part of the same event. A body begins with one unit and receives five additional units per census. The specified damaged body spends three on repair, one on care, and reserves two for possible outputs. The model accounts for the complete six-unit budget. Outputs inherit the restored code before undergoing the specified later damage. The claim is consequently stronger than an interface that simply assumes successful repair, available funds, and faithful inheritance independently. Within this source, those relations are tied to the same body and transaction.

It remains a finite mathematical model. Its physical cells and payments are explicit model variables, not observations of a constructed device. The bootstrap and writing operations are trusted primitives. The damage leaves information in the backups. The external resource inflow is prescribed. Later noise can still eliminate the body. These qualifications determine what the example shows: a logically and causally specified conjunction of repair, execution, and reproduction is possible. It does not show that every organism has this architecture or that an arbitrary damaged system can recreate information it has entirely lost.

The accompanying failures make the positive claim informative. Cutting the bootstrap prevents the specified repair. Insufficient funds at the repair stage prevent the required paid operation. Erasing all programme information cannot leave a single decoder that recovers two distinct original programmes from the same remaining input (UEOT Project, 2026c). These are failures of different enabling conditions. None is repaired by observing that some abstract controller could output the desired bit pattern. The problem is whether this body, with these accessible resources and operations, can do so.

Here the preservation condition changes the explanation. In the named raw-execution witness, two bodies have the same decoded instruction: the open gate permits the specified bloom action, while the closed gate blocks raw execution. This comparison fixes the diagnostic argument; it is distinct from computing a new diagnosis or executing a complete repair transaction. Code identity alone therefore does not determine executable control. Two post-repair bodies can have identical birth records while differing in the paid history that made those records possible. Output identity alone does not recover that repair history. An explanation of this recovery has to retain the causal route, even when prediction of later outputs does not.

4.3 Continuity, copying, and branching

A repair story also contains a claim about whose activity continued. The source constructs retention and strict birth as different events, assigns their corresponding identities, and records parentage. The labels track that event convention; names alone do not establish causal continuity. A retained body and a newborn carrying the same programme are not the same token merely because they behave alike. The historical record permits questions about continued identity, lineage, and reproducibility to have different answers. This is a mathematical implementation of a distinction; it does not prove that its chosen identity convention is the only legitimate one for natural organisms.

Process philosophy already treats material replacement and organized continuity as central to biological persistence (Dupré and Nicholson, 2018). The present point is that a representation can preserve a process's output organization while erasing the history required by a particular continuity claim. Restoring the same organization from a saved description might establish reproducibility. Showing that a damaged bearer carried out the restoring operation might establish continuity under an appropriate tracking relation. Neither follows from matching future distributions. A causal-historical bridge needs to specify which relations connect the earlier and later bearer and why those relations are relevant to the inquiry.

Branching exposes the limit of a purely causal criterion. Suppose one organization divides into two equally continuous successors. Both can inherit its structure, memory, and activity. If both were numerically identical to the predecessor, the symmetry and transitivity of identity would make them identical to one another, contrary to their stipulated distinctness. A lineage relation can branch; numerical identity cannot do that same work without revision. The alternatives include selecting a privileged continuation, treating the original as ending in division, or acknowledging that the inquiry uses a different persistence relation. Nothing in predictive equivalence selects among them.

This is especially important for teleology. A reproductive activity can preserve an organization through new tokens while ending the individual that performed it. A purpose explanation directed at lineage continuation can be true without being an explanation of that individual's continued existence. Recent discussion of organizational teleology rightly treats diachronic identity as part of the naturalization problem (Cuciniello, 2026). The contribution here is to make the relevant loss of identity information visible in the same source used to discuss growth, rather than to claim that causation alone solves every identity puzzle.

The resulting realism is limited but substantive. It permits recognition of organized processes whose continuity is supported by causal history, while allowing that some cases remain indeterminate or depend on a stated tracking convention. It rejects an inference from behavioral resemblance to numerical identity and an inference from causal descent to uninterrupted individual survival. Those rejections become decisive when a theory attempts to explain maintenance or purpose using a representation that has discarded the bearer of the relevant history.

5 The grounds and limits of purposive explanation

5.1 What an objective does not explain

An observed action sequence can often be made optimal by assigning it a suitable reward. That possibility does not explain why the system acts as it does or what its activity is for. The inverse reinforcement learning literature already identifies underdetermination of reward by optimal behavior (Ng and Russell, 2000, §3.1). The current UEOT purpose counterexamples make a related, restricted point: bare evidence and feasible dynamics do not select a unique externally attached evaluation. One formal construction simply pairs the same evidence value with different objective functions. It must not be inflated into a proof that two completely identical physical mechanisms have incompatible intrinsic purposes (Qian, 2026a, §18.3, N-TEL-01; UEOT Project, 2026c, supplementary matrix F21).

The distinction is between evidence for an evaluation and the mere availability of a mathematical score. A regulatory target can have support from the sensor-controller relation and the system's responses to perturbations. A selected function can have support from the history explaining a trait's retention. A designed function can have support from a construction history. A contribution to a capacity can be established by causal analysis of the organized system. These are different explanatory achievements, as the contrast between etiological and causal-role accounts of function makes clear (Wright, 1973, 156-159; Cummins, 1975, §III.3). A causal contribution alone need not be purposive; a further design, historical, or regulatory ground has to support that attribution. Calling all these achievements optimality suppresses the evidence needed to distinguish them.

Independence here is evidential, not spatial or causal. A controller's target need not come from outside the system, and a living organization need not contain an explicit utility register. The requirement is that the evidence for the relevant distinction does not consist only in fitting a favourable score to the be-

havior to be explained. A directly investigated repair mechanism may support the claim that its activity restores a working organization even when several numerical rewards rationalize that activity. Conversely, an externally chosen population score may be precisely calculable without being represented as a goal by any individual.

The finite source contains different kinds of relation. Its majority routine restores damaged instruction cells under specified perturbations. This is an implemented capacity. Calling restoration its function has an additional ground in this designed model: the stated construction task is to recover the pre-damage programme, whose information is retained in the copies, and the routine is organized for that task. This standard is not inferred from whichever output happens to occur. It licenses a designed-function description, not an intrinsic biological norm. The growth criterion instead evaluates population consequences over time. Attainment does not show that the bearer anticipates growth, values it, or regulates an estimate of it. The model establishes implemented recovery and demographic performance; it does not make the latter an intrinsic intention.

5.2 Growth from the same transactions

Let s be the retention probability under care, b the birth probability under care, δ the retention cost of bloom, γ its birth advantage, and q the probability that an output is damaged. In the currently classified domain,

$$0 < \delta < s < 1, \qquad b > 0, \quad \gamma > 0, \quad b + \gamma < 1, \qquad 0 < q < 1.$$

The expected total outputs of the two executable actions are

$$k_c = s + b, \qquad k_b = s - \delta + b + \gamma.$$

The hereditary programmes introduced in Section 3 have exponential factors

$$\lambda_0 = k_c, \qquad \lambda_1 = (1 - q)k_b + qk_c.$$

These expressions come from the survival and birth transactions, the output damage distribution, and the programme executed by each diagnostic type. They are not rewards inferred from the observation that a chosen programme wins. The distinction between retention and strict birth remains in the actual histories even when the total population count adds them (Qian, 2026a; UEOT Project, 2026a).

Under the source's full permitted public-history policy class and canonical finite nonempty initial conditions, the least positive common exponential upper factor for expected population is

$$\rho_* = \max\{1, \lambda_0, \lambda_1\}.$$

The exact result is expressed by bounds over every finite census. For each fixed parameter choice and fixed nonempty finite canonical initial population, there is a constant C, independent of policy and census time, such that $\mathbb{E}_\pi N_t \le C\rho_*^t$ for every permitted policy π and every finite t. One specified stationary policy has positive upper and lower constant multiples of the same exponential. No smaller positive base supplies a common bound of that form. This does not assume that every competitor has a growth limit. The corresponding positive-root and limsup statements are analytical consequences of these finite bounds, not separate new formal limit endpoints.

The two internal programmes attain precisely under the following conditions:

$$\theta_0 \text{ attains } \rho_* \iff \gamma \le \delta \text{ and } k_c \ge 1,$$
$$\theta_1 \text{ attains } \rho_* \iff \gamma \ge \delta \text{ and } \lambda_1 \ge 1.$$

Each result is connected to that programme's own damaged founder, its paid restoration, and its inherited execution. This is the positive achievement of using one source. Repair is not demonstrated in one model and optimality in an unrelated model with more convenient resources. Nevertheless, the same-source connection does not erase the evaluation choice: expected population is still the specified quantity being compared.

The term 1 in ρ_* reveals why implementation cannot be treated as a harmless technical detail. A damaged body's bloom request would cost more than its available six units once repair and output reserve are included. The actual transaction rejects that request, preserves the body, returns the incoming resources, and does not expose it to that transaction's noise. Repeating rejection therefore supplies a population factor of one. This is a declared feature of this finite source, not a natural law that inactivity prevents decay. If rejection instead exposed the body to an independent mortality process, the action rows and resulting optimization problem would change.

The permitted controller can use the public history and jointly randomized requests; it cannot inspect the current world noise before choosing or use hidden audit fields as observations. The comparison is also supported by continuing resource inflows. It is not a theorem about indefinite growth from one closed finite wallet. These premises identify the optimization problem to which the conclusion belongs. Enlarging the information set or replacing the rejected transaction with a different physical event would change the problem rather than strengthen its answer.

5.3 Which persistence is favoured?

On a healthy body, bloom decreases the original individual's retention probability from s to $s-\delta$ while increasing birth probability from b to $b+\gamma$. Where the gain is sufficiently large, the population criterion favours bloom. The event can thus be growth-promoting and individually more dangerous at the same time. A statement that the action promotes persistence acquires a determinate truth value only after its bearer and measure have been specified. Population increase does not retrospectively make the dead original body survive.

Even within the population evaluation, an exponential optimum differs from the best count at a deadline. At the classified parameter point

$$(s, \delta, b, \gamma, q) = (2/3, 1/8, 1/4, 1/4, 1/3),$$

$\lambda_1 = \rho_* = 1$. From the damaged founder, the internal θ_1 programme has expected population $11/12$ at every positive census. A policy that repeatedly makes the rejected damaged-body request retains one body on every path. Both have exponential factor one, although their finite-time expectations differ (Qian, 2026a; UEOT Project, 2026a). The comparison shows why an asymptotic ranking cannot silently substitute for every practical success criterion.

Nor does a higher expected factor imply that the preferred programme survives in each finite realization. The source includes an exact finite-population event in which the higher-growth programme disappears while the lower-growth programme remains. Its selection theorem retains the narrower original regime: the probability conditions above together with $\gamma > \delta$ and $s + b > 1$. For each specified

share tolerance $0 < \varepsilon \leq 1$ and error probability $\eta > 0$, it chooses a finite positive horizon and then a finite number of independent cohorts. Each cohort begins with one healthy founder of each programme and has its own continuing resource account. At the chosen census, the high-programme share is at least $1 - \varepsilon$ with probability at least $1 - \eta$, counting extinction as failure. Independence concerns the cohort noise laws, not funding alone. This is not almost-sure selection over unlimited time in one fixed finite population (UEOT Project, 2026c, supplementary matrix F18-F19). Selection, implemented action, and optimality therefore have separate evidential burdens.

These examples support pluralism about purposive explanations, but not unrestricted subjectivism. The survival probability, birth rate, payment, and genealogy are fixed by the model. The evaluator cannot reverse their values by changing a label. What remains open is which relation grounds a particular purpose claim and which entity bears it. Restoring the individual, transmitting a programme, increasing expected descendants, and retaining a selected trait can all be naturalistically describable without constituting one universal end.

5.4 Functional standards and reasons

Organizational accounts seek more than an observer's preference by connecting functional contributions with the conditions that maintain the organization. Mossio, Saborido, and Moreno give a substantial account of such functions and their normative dimension (Mossio et al., 2009). Bich's regulatory development asks how activities are coordinated in ways that matter to the organism's continued organization (Bich, 2024). These are genuine competitors to a view that only externally assigned scores make teleology intelligible. The present paper accepts that organization and regulation can supply evidence unavailable from behavior fitting alone.

The force of the resulting standard remains contested. Corti's objection concerns biological functional correctness: why does dependence of an organization's existence on an activity establish how a component ought to work, including a basis for malfunction? It is not merely a question about moral goodness (Corti, 2023, §3). Causal effectiveness and counterfactual dependence do not by themselves answer it. The present model's repair standard is licensed by its specified design task and the relation to the previously encoded programme. That warrants success or failure relative to a designed function. It does not establish that this standard is an intrinsic biological norm grounded only in the organization's continued existence.

Four levels must consequently remain distinct: a descriptive causal contribution; a functional attribution supported by design, formation history, or an adequately defended regulatory account; intrinsic biological normativity; and moral or practical reasons. The model securely demonstrates the first and supports a designed-function instance of the second. It does not resolve the third. Even if a stronger biological account supplied that further ground, justification of moral demands would remain an additional issue. This leaves the strongest normativity objection open at its proper level instead of pretending to answer it with a restriction on ethics.

The positive purposive claim is therefore specific: under the independently identified design standard, the recovery routine is for restoring executable code. The population calculation is a demographic performance explanation unless further grounds make growth a function or target of the bearer. Its precision does not itself supply those grounds. Neither conclusion attributes consciousness. Autonomous agency would also require further arguments concerning individuality, interaction, and normativity of the kind already developed in the agency literature (Barandiaran et al., 2009). The distinction preserves

a real implemented accomplishment without turning every causally effective process into a purposive one.

6 What is unified, and what remains plural

6.1 A shared constraint on explanatory transfer

The two main arguments can now be joined. Predictive identification relates histories through selected future records. Persistence relates earlier and later bearers through an appropriate continuity claim. Purposive explanation relates an activity to an outcome under a function, control, or historical account. These relations need not have the same extensions. A body can retain a predictive type after a costly recovery; a newborn can share that type without being the earlier body; a programme can increase descendants while diminishing its bearer's retention. The unification proposed here consists in making these differences constrain explanations of the same system.

Consider a constructed inference that tests the original UEOT ambition to connect persistence, optimization, and purpose: the predictive organization survives, the surviving organization optimizes growth, therefore the object's repair is for its own optimal persistence. This is a diagnostic reconstruction, not a quotation or an allegation that computational mechanics or organizational teleology makes this inference. Each clause may be defensible under some reading. The birth quotient preserves the predicted organization; a policy attains the growth factor; an individual undergoes repair. Yet the quotient may have erased the repair expenditure, the optimal policy may be externally specified rather than internally executed, and the counted population may contain new identities. Persistence changes its bearer while optimality changes its standard. The conclusion is not supplied by placing the separately true premises beside each other.

The successful alternative has greater content than a warning. Let $R(x)$ mean that admissible system x performs the specified paid recovery, and $O(x)$ mean that it attains the stated benchmark. Separate claims $\exists x\, R(x)$ and $\exists y\, O(y)$ do not imply $\exists z\, [R(z) \wedge O(z)]$. The source supplies the latter conjunction for each named programme in its attainment region. The common witness fixes the damaged founder, programme, backups, bootstrap, transaction account, information permissions, and subsequent output process. Repair and attainment are evaluated on that common realization.

The source remains constructed, and its resource inflow and rejection rule remain modeling choices. Its added content is not that nature must instantiate them. It is that the already specified six-unit account and inherited execution impose potentially conflicting demands: damaged bloom cannot be funded, repair alone does not ensure attainment, and an external policy need not coincide with the stored programme. The result resolves those demands under explicit parameter conditions without changing the source for each desired property. This is a conditional compatibility result; its empirical application still requires evidence for the source assumptions.

This conjunction still warrants only the purpose claim supported by its ground. It establishes restoration and growth attainment, not an internal representation of growth. It establishes a causal relation between programme and outcomes, not a moral reason to maximize them. Thus the preservation condition joins the two arguments without dissolving their differences. Object identification constrains whose activity is explained; purpose attribution constrains which outcomes and alternatives the representation must retain. Each can expose a deficiency in a representation accepted for the other.

6.2 Relation to the closest alternatives

The position continues a cybernetic insight that purpose can be approached through organized relations among behavior, feedback, and outcomes (Rosenblueth et al., 1943). It also shares interventionism's emphasis on the range over which explanatory relations remain stable under changes (Woodward, 2000). Its distinctive emphasis is the compatibility of those relations with claims about the bearer's identity. An input-output equivalence can be satisfactory until the explanation asks which material or historical organization realizes the response. A controller can be accurately described until its apparent objective depends on an action it cannot perform.

Organizational teleology presents the closest challenge. Bich's account concerns regulation that coordinates activities in relation to an organism's organization, not just the presence of a mechanism (Bich, 2024, §§4-5). Applying that criterion to the device is an interpretation offered here, not a verdict attributed to Bich. Diagnosis and the execution gate provide state-sensitive control, but this does not demonstrate the wider biological organization or endogenous normative ground his account seeks. External resources and trusted components are not automatically disqualifying; their place in the claimed organization still needs explanation. The preservation analysis adds a specific test: a representation erasing the gate cannot preserve this raw-execution distinction, even if it retains the decoded code. It assesses a representation used by a regulatory explanation without claiming to replace that explanation.

Cuciniello distinguishes the achievements and difficulties of minimal and regulative organizational teleology, including the problem of how regulatory goals and changing organizations are grounded (Cuciniello, 2026, §§4-6). The present fixed-code recovery model does not explain how novel goals arise or how an organization transforms its own standards. Its separate retention and lineage records can instead identify which bearer a proposed developmental or reproductive explanation concerns. This is a narrower addition: an account that moves from restored individual organization to transmission across generations must justify that change of bearer. Neither competitor is accused of neglecting mechanisms or history. Their stronger naturalization questions remain open here; the proposed contribution is a way to test whether a representation preserves the relations their explanations invoke.

Other unifying programmes illustrate why scope matters. The free-energy principle proposes a broad relation among organismic maintenance, inference, and action; technical criticism has questioned particular derivations and their assumptions (Friston, 2010; Biehl et al., 2021). This article neither identifies its population objective with variational free energy nor treats criticism of one formulation as a refutation of every version. The comparable question is which bridge connects a mathematical objective, a concrete generative process, and an implemented policy. Naming the same functional across levels does not itself establish that connection.

Likewise, results about causal emergence compare micro and macro descriptions under a specified intervention distribution and information measure (Hoel et al., 2013). They do not settle that every coarser description possesses greater causal power. They reinforce the need to state what a comparison measures. The present view allows multiple useful scales while requiring a representation to preserve the causal differences on which the particular explanation depends.

This form of unity is methodological and philosophical. It does not derive quantum dynamics, biological autonomy, and social institutions from a shared equation. In each domain, the realization of an organization, the legitimacy of interventions, and the ground of an evaluative distinction require further inquiry. UEOT's current mathematical closure is closure of declared conditional claims and their boundaries (UEOT Project, 2026d). Its current physics work retains domain-specific assumptions

and unresolved empirical responsibilities (UEOT Project, 2026b). The paper's proposed unity can be assessed without accepting a completed universal natural theory.

7 Objections and consequences

7.1 The realist step remains contestable

An empiricist can accept every mathematical result in this paper while declining its realism. The finite source is a stipulated model; proving relations within it does not show that nature contains the organizations it depicts. Even if a natural system were accurately modeled, predictive and intervention success might be treated as grounds for accepting the model's observable consequences rather than its ontology. This objection identifies a real limit. The argument does not deductively force realism on an opponent who rejects explanatory inference to unobserved organization.

The positive case is comparative. If a proposed organization explains both ordinary behavior and changes following the loss of a gate, backup, or resource, while a rival preserving only ordinary outputs does not, there is a reason to treat the organization as more than a convenient summary of that output record. The same relations account for independently varying effects. The inference is defeasible because another realization could explain them, the intervention may have additional effects, or the purported continuity may be misidentified. Realism here means accepting the explanatory organization to the extent justified by those relations, with revision when their support fails.

This commitment is more limited than an inventory of intrinsically bounded substances, but it is not merely a renaming of prediction. It recognizes causal organization and historical dependence as possible features of the world that a predictor can successfully ignore. A purely predictive type can be an excellent instrument without being the complete bearer of the explanation. The paper would lose its philosophical point if it treated all representations as interchangeable whenever their ordinary observations matched.

7.2 Preservation can look tautological

The preservation condition may appear to say only that an explanation must retain what it needs. Taken alone, it does. Its value cannot lie in the elementary fibre argument or in the introduction of another abstract criterion. The substantive work is determining which distinctions a seemingly successful explanation has actually erased, why they matter to its conclusion, and whether retaining them is compatible with the other desired claims.

The paired examples supply that work. Complete birth prediction does not require the initial repair fee, although reconstructing this bearer's paid-repair history does. Code equality does not require equality of execution gates, although recovering the specified raw-execution distinction does. Expected population combines retained and new individuals, although a continuity explanation distinguishes them. Growth attainment does not preserve finite-deadline rankings, although a decision concerning a deadline requires them. These omissions are easy to miss because the representation is successful in each of its original tasks. The argument identifies a systematic route from legitimate compression to illegitimate transfer.

The positive source construction adds a non-vacuity requirement. It is insufficient to demand all desirable properties if they cannot coexist. In the stated parameter regions, the paid repair, inherited internal execution, and population attainment can coexist. Outside those regions, failure of the two

named programmes does not rule out every possible repairing strategy; a permitted policy can, for example, pay for an initial repair and then follow another rule. The result is an assessment of particular connections, not a declaration that any bridge one wants can be added. A reader may still judge this to be methodological synthesis rather than a new metaphysics. That is a defensible classification of the contribution, provided the concrete explanatory discriminations are acknowledged.

7.3 A model built to repair is a weak case for natural purpose

The strongest objection to the purposive reading is that a deliberately constructed error-correcting model displays the investigator's design, not the origins of natural purposiveness. The bootstrap, preserved backups, resource inflow, and programme structure already encode many conditions of success. A construction that starts with them cannot explain how an organism acquires them or why its organization constitutes a norm for itself. The present argument accepts this limitation.

What the model establishes is more local: the presence of enabling conditions does not make the implemented recovery fictitious. Restoration actually changes damaged execution conditions, and the source identifies failures when the necessary means are lost. A system can perform some of its own maintenance while depending on supports it does not produce. Treating every dependency as disqualifying self-maintenance would make the relevant contrast disappear, since the question concerns which conditions the organization itself restores. Treating that limited restoration as complete autonomy would be the opposite mistake.

A naturalized attribution therefore requires an additional account of how the restoring organization is established and sustained in the target. Selection history, developmental organization, or ongoing regulation may provide different grounds. Their adequacy cannot be read from a favourable expected-growth number. Nor must all purposive explanations await one universal theory of intrinsic value: a causal-role explanation can succeed on its stated terms while an account of autonomous normativity remains disputed. The paper defends that limited success and the need to identify which further claim is being attempted.

7.4 The framework risks conflating unlike systems

If every persistent organization is admitted, a rock, a thermostat, a repaired register, and an organism may seem to become equivalent examples of purpose. The objection would be decisive if the account inferred agency from persistence. It does not. Enduring within a domain establishes neither an error-sensitive response nor an internally implemented repair. Regulation establishes neither production of the regulating components nor a selection history. Reproduction establishes neither conscious intention nor justified action. The relations can overlap without forming a single ascending scale on which each system has one score.

The finite device is consequently evidence for a restricted compatibility claim, not a definition of life. A passive object can be real without pursuing a goal. An organism can have reproductive functions whose effects conflict with its immediate survival. An agent can deliberately undertake a finite activity while accepting that it will not preserve either itself or its organization indefinitely. The vocabulary should track those differences rather than declare them deficient approximations to one optimum.

The same restraint matters for human meaning. Understanding the material and social conditions of a commitment can explain what carrying it out requires and who bears its costs. It does not show that the commitment is justified because it maintains a larger system. A reason to oppose a destructive

institution need not be invalidated by the institution's capacity to reproduce itself. The reconstructed UEOT can illuminate the feasibility and consequences of commitments while leaving their justification to arguments about persons, harms, obligations, and alternatives. That boundary preserves explanatory continuity without deriving ethics from demographic success.

7.5 The discriminating consequences are real but not yet empirical findings

A philosophical framework should change how a claim can fail. This one supplies several discriminations. If two models fit the same birth record but disagree about repair expenditure, birth data alone cannot choose between their resource explanations. If restoring output requires an external overwrite after the internal bootstrap is removed, the result does not confirm the original claim of internally implemented recovery. If a policy succeeds only by inspecting current noise, it does not establish success within the original information restriction. If a tracking scheme counts a replacement as a retained token, its population totals cannot independently verify uninterrupted individual persistence.

An empirical application could investigate these differences in a device with inspectable memory and controlled faults. Its output task, internal restoration operations, energy account, and identity tracking would first need independent specification. One could then compare ordinary output fit with the effects of disabling the repair path, modifying accessible resources, or changing the allowed action interface. A recovered output would be evidence for different claims depending on whether it came from internal correction, external replacement, or a previously hidden backup. This is a proposed research design illustrating the argument, not an experiment performed for this article.

Biological application would be harder because interventions can damage several capacities at once and boundaries change during development. The difficulty does not license pretending that a successful abstract fit resolves the realization question. It identifies why evidence for regulation, organizational continuity, and function often has to be assembled from several sources. A framework has empirical relevance when it clarifies which observations would distinguish competing explanations, even if its present contribution is to logical and conceptual evaluation rather than a new measured effect.

The elementary factorization condition is analytic; its logical opposite is not a useful empirical falsifier. What can be wrong is the judgment that an explanation requires a certain distinction, that the chosen representation actually preserves it, or that a natural process realizes the proposed mechanism. The realist interpretation and the grounds of a purpose attribution are also contestable. Beyond those failures, the synthesis may add too little to established accounts to merit a distinctive philosophical position. Formal precision helps expose these questions but does not settle them. Their assessment requires comparison with alternatives and criticism from readers outside the project.

8 Conclusion

The revised UEOT can sustain a philosophical position without a universal topological loop, a unique optimum, or a derivation of obligation from persistence. Organized processes can be real bearers of causal and historical explanation even when their best predictive descriptions depend on the task. Implemented restoration establishes a causal capacity; an independently supported design, historical, or regulatory account can additionally license an appropriate purpose attribution. The finite example supplies a designed restoration function and a separate demographic performance result, leaving intrinsic biological normativity unresolved.

The two claims are connected by the distinctions an explanation must preserve. In the finite source, predictive equivalence can erase paid recovery, decoded information can erase executable control, and population growth can erase the fate of the original individual. The positive construction shows how some of these relations can instead be satisfied together, under explicit assumptions, by the same damaged founder and its successors. That conjunction establishes compatibility and disciplined transfer, not a universal natural purpose.

The resulting unity is a commitment to explain the same organization across descriptions while allowing different explanations to answer different questions. Its philosophical merit lies in making the passage between those questions answerable to causal realization, history, and the grounds of evaluation. Its remaining ambition is to improve the explanation of particular systems, and to let those explanations decide how far the unity extends.

Sources, evidence, and assistance

This article uses the historical UEOT paper and the maintained philosophy and mathematics books at repository commit `3d443ec29eb17af1fc00699a74e2d42d73516d81`, inspected on 11 November 2026. Mathematical references distinguish written arguments, existing formal endpoints, and source-matched archived compilation evidence. No new Lean theorem, complete Lean rebuild, physical experiment, or empirical validation was performed for this article. The complete task-dependent 2/3/4 classification and the local rank-three response example are written mathematical results with exact source checks, not newly supplied dedicated formal endpoints.

The accompanying formal claim-evidence matrix gives exact files, endpoint names, assumptions, and exclusions. Its source routes are F01-F02 for predictive minimality and the common conditional source; F03-F07 for invariant limiting structures, homology, failure families, and access; F08-F10 for execution, tag projection, and the birth classifications; F11-F12 for paid repair and its failures; F13-F17 for growth and internal attainment; F18-F20 for selection and marked identity; F21 for purpose nonidentification; F22-F26 for tied optimization, limiting structures, and the dual-drive boundaries; and F27 for current scientific status. The general claim matrix separately marks this article's philosophical interpretations. These supplements belong with the article when its technical support is assessed.

Degui Qian is the author of the original UEOT programme and the named author of this manuscript. ChatGPT assisted with research, drafting, literature checking, and technical preparation. One coordinating assistant wrote the manuscript; three native research agents separately audited historical and philosophical claims, mathematical scope, and prior art, and reviewed the complete draft. These are internal AI-assisted reviews, not independent human peer review. No external reviewer endorsement is asserted. The author remains responsible for the scholarly claims and any submission decision.

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